SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE: CSA13**

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

Analytical Problem

Code of Conduct:

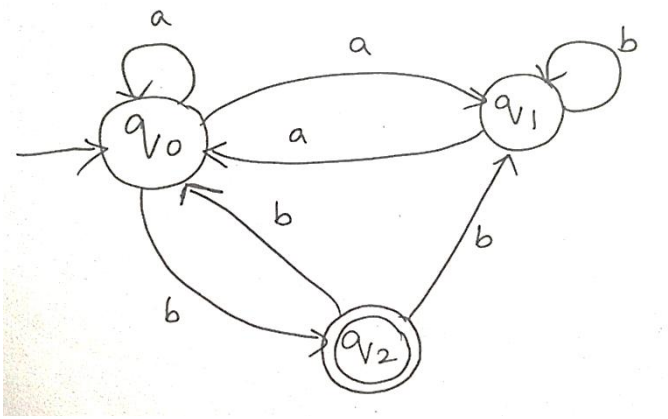
I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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1.Consider the NFA given below which accepts some language L over the alphabet **Σ**={a,b}. Construct a DFA that represents the same language L

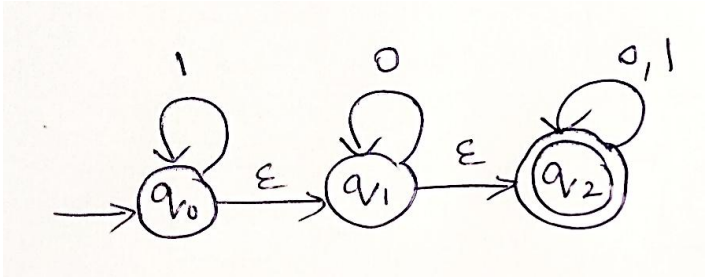


2.Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet **Σ**={0,1}. Construct a DFA that represents the same language.

5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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ASSESSMENT TOOL - ANALYTICAL PROBLEM

Course Name: Theory of Computation	**Course Code:** CSA1308
Name: D.Naveen	**Reg No:** 192372177
CO1: Analyze fundamental mathematical concepts of automata theory (BL:3)	**Marks:** 25 | **Duration:** 1 Hour

Code of Conduct: I, **D.Naveen (192372177)**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Question 1

Consider the NFA given below which accepts some language L over Σ = {a, b}. Construct a DFA that represents the same language L.

Step 1: Transition function read from the given NFA

Start state = q_0, Final state = q_2.

State	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$* q_2$	Φ	$\{q_0, q_1\}$

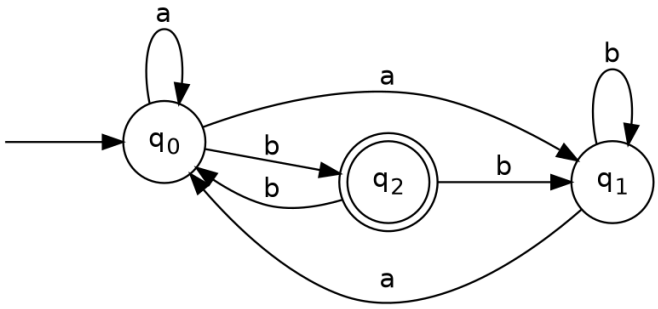


Fig. 1.1 - Given NFA

Step 2: Subset construction

- $\{q_0\}$: on a $\rightarrow \{q_0, q_1\}$; on b $\rightarrow \{q_2\}$
- $\{q_0,q_1\}$: on a $\rightarrow \{q_0,q_1\} \cup \{q_0\} = \{q_0,q_1\}$; on b $\rightarrow \{q_2\} \cup \{q_1\} = \{q_1,q_2\}$
- $\{q_2\}$: on a $\rightarrow \Phi$; on b $\rightarrow \{q_0,q_1\}$
- $\{q_1,q_2\}$: on a $\rightarrow \{q_0\} \cup \Phi = \{q_0\}$; on b $\rightarrow \{q_1\} \cup \{q_0,q_1\} = \{q_0,q_1\}$
- Φ (dead state): on a, b $\rightarrow \Phi$

No new states appear, so the construction stops. Any subset containing q_2 is a final state.

Step 3: DFA transition table

DFA state	a	b
→ A = $\{q_0\}$	B	C
B = $\{q_0,q_1\}$	B	D
* C = $\{q_2\}$	Φ	B
* D = $\{q_1,q_2\}$	A	B
Φ (dead)	Φ	Φ

Step 4: Equivalent DFA

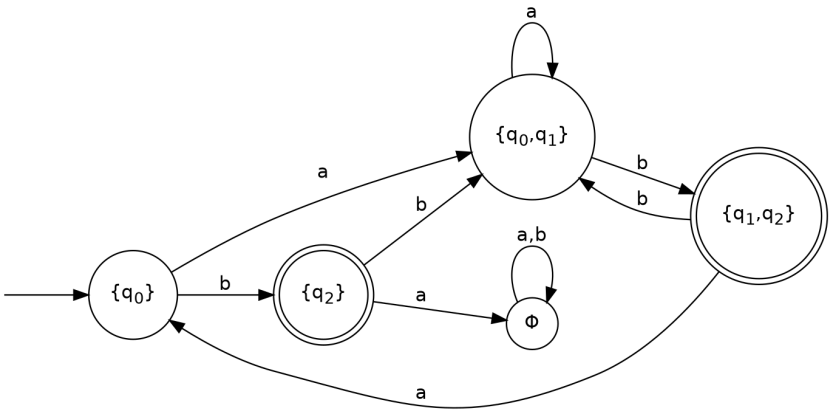


Fig. 1.2 - DFA for L (final states: $\{q_2\}$ and $\{q_1,q_2\}$)

Question 2

Let L be the language of words of length at least 2 whose last two letters are different from each other (e.g. abaab $\in$ L but bb $\notin$ L). Construct an NFA for L, then draw the equivalent DFA.

Step 1: NFA

Idea: stay in q_0 reading any prefix, then guess that the last two symbols begin. Either "a then b" or "b then a" must be read, after which the input must end.

- q_0 loops on a, b (skips the prefix).
- $q_0 \to q_1$ on a, then $q_1 \to q_3$ on b (ends with "ab").
- $q_0 \to q_2$ on b, then $q_2 \to q_3$ on a (ends with "ba").
- q_3 is the only final state.

State	a	b
$\to q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	Φ	$\{q_3\}$
q_2	$\{q_3\}$	Φ
$* q_3$	Φ	Φ

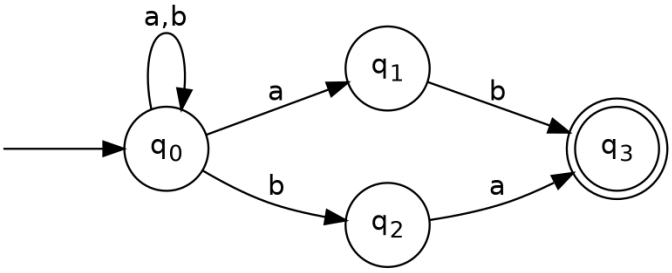


Fig. 2.1 - NFA for L

Step 2: Subset construction

- A = $\{q_0\}$: a $\to \{q_0,q_1\}$ = B; b $\to \{q_0,q_2\}$ = C
- B = $\{q_0,q_1\}$: a $\to \{q_0,q_1\}$ = B; b $\to \{q_0,q_2,q_3\}$ = D
- C = $\{q_0,q_2\}$: a $\to \{q_0,q_1,q_3\}$ = E; b $\to \{q_0,q_2\}$ = C
- D = $\{q_0,q_2,q_3\}$: a $\to \{q_0,q_1,q_3\}$ = E; b $\to \{q_0,q_2\}$ = C
- E = $\{q_0,q_1,q_3\}$: a $\to \{q_0,q_1\}$ = B; b $\to \{q_0,q_2,q_3\}$ = D

States D and E contain q_3, so they are final.

Step 3: DFA transition table

DFA state	a	b
$\to$ A = $\{q_0\}$	B	C
B = $\{q_0,q_1\}$	B	D
C = $\{q_0,q_2\}$	E	C
* D = $\{q_0,q_2,q_3\}$	E	C
* E = $\{q_0,q_1,q_3\}$	B	D

Step 4: Equivalent DFA

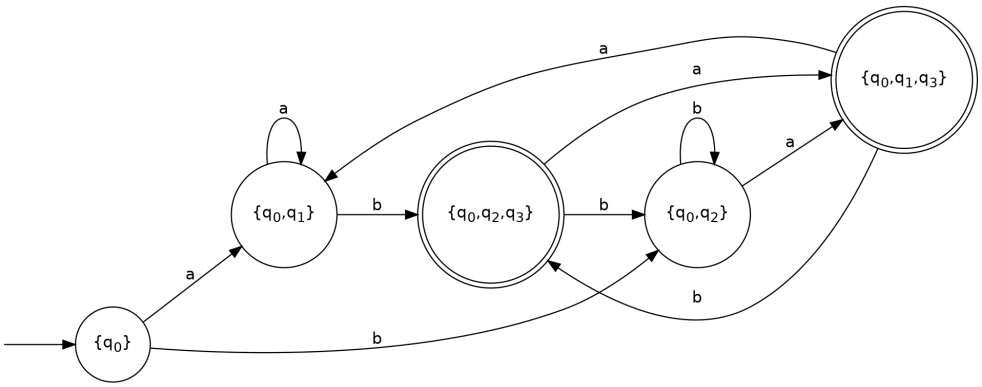


Fig. 2.2 - DFA for L

Check: abaab: A -a-> B -b-> D -a-> E -a-> B -b-> D (final), so it is accepted. bb: A -b-> C -b-> C (not final), so it is rejected.

Question 3

Design an NFA that takes a binary string as input and accepts the string only if the last two digits are the same (00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted.

Design idea

Loop in q_0 on any symbol, then non-deterministically guess that the final two symbols have started: "00" goes through q_1 to q_3, and "11" goes through q_2 to q_3.

Formal definition

NFA M = (Q, Σ, δ, q_0, F) where

- Q = {q_0, q_1, q_2, q_3}
- Σ = {0, 1}
- Start state q_0
- F = {q_3}
- δ: Q × Σ → 2^Q is given below:

State	0	1
→ q_0	{q_0, q_1}	{q_0, q_2}
q_1	{q_3}	Φ
q_2	Φ	{q_3}
* q_3	Φ	Φ

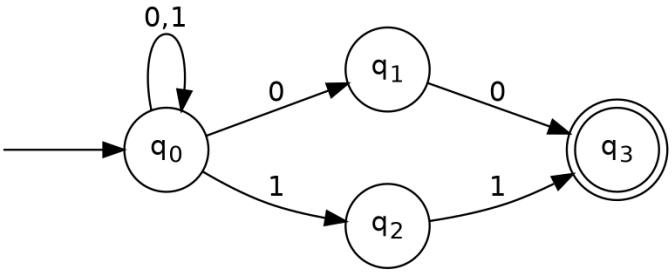


Fig. 3.1 - NFA accepting strings ending in 00 or 11

Acceptance of the string 01011

Input: 0 1 0 1 1. One accepting path exists:

Symbol read	Current state	Next state
(start)	-	q_0
0	q_0	q_0
1	q_0	q_0
0	q_0	q_0
1	q_0	q_2 (guess: last two = 11)
1	q_2	q_3

Computation: q_0 -0-> q_0 -1-> q_0 -0-> q_0 -1-> q_2 -1-> q_3
The whole input is consumed and the machine is in $q_3 \in$ F. Hence **01011 is accepted** (δ*(q_0, 01011) contains q_3).

Question 4

Consider the NFA which accepts the language of all strings having 001 as a substring over Σ = {0, 1}. Construct a DFA that represents the same language.

The NFA diagram was not included in the question paper, so the standard NFA for "contains 001" is used.

Step 1: NFA

State	0	1
→ q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	Φ
q_2	Φ	$\{q_3\}$
* q_3	$\{q_3\}$	$\{q_3\}$

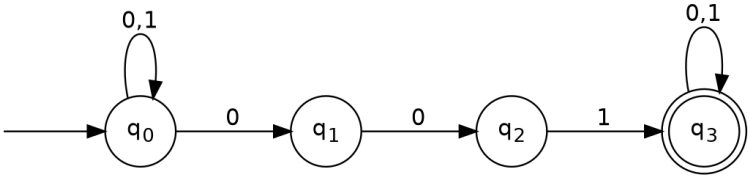


Fig. 4.1 - NFA for strings containing 001

Step 2: Subset construction

- A = $\{q_0\}$: 0 → $\{q_0,q_1\}$ = B; 1 → $\{q_0\}$ = A
- B = $\{q_0,q_1\}$: 0 → $\{q_0,q_1,q_2\}$ = C; 1 → $\{q_0\}$ = A
- C = $\{q_0,q_1,q_2\}$: 0 → $\{q_0,q_1,q_2\}$ = C; 1 → $\{q_0,q_3\}$ = D
- D = $\{q_0,q_3\}$: 0 → $\{q_0,q_1,q_3\}$ = E; 1 → $\{q_0,q_3\}$ = D
- E = $\{q_0,q_1,q_3\}$: 0 → $\{q_0,q_1,q_2,q_3\}$ = F; 1 → $\{q_0,q_3\}$ = D
- F = $\{q_0,q_1,q_2,q_3\}$: 0 → $\{q_0,q_1,q_2,q_3\}$ = F; 1 → $\{q_0,q_3\}$ = D

D, E, F contain q_3, so they are final (they are equivalent and can be merged, giving a minimal 4-state DFA).

Step 3: DFA transition table

DFA state	0	1
→ A = $\{q_0\}$	B	A
B = $\{q_0,q_1\}$	C	A
C = $\{q_0,q_1,q_2\}$	C	D
* D = $\{q_0,q_3\}$	E	D
* E = $\{q_0,q_1,q_3\}$	F	D
* F = $\{q_0,q_1,q_2,q_3\}$	F	D

Step 4: Equivalent DFA

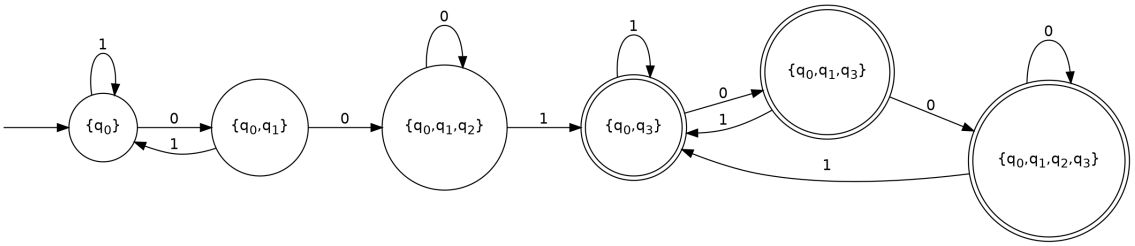


Fig. 4.2 - DFA for strings containing 001

Question 5

An NFA with ε-transitions is given. Find the ε-closure of each state and design an equivalent DFA. Draw its transition table and mark the initial and final states.

Given ε-NFA

State	0	1	ε
→ q_0	Φ	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	Φ	$\{q_2\}$
* q_2	$\{q_2\}$	$\{q_2\}$	Φ

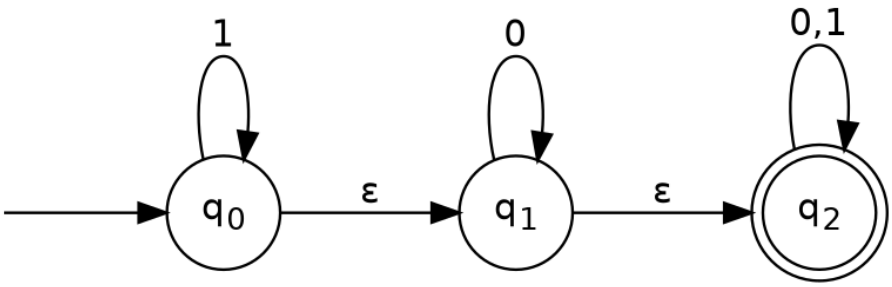


Fig. 5.1 - Given ε-NFA

Step 1: ε-closures

- ε-closure(q_0) = $\{q_0, q_1, q_2\}$ ($q_0 \to_\varepsilon q_1 \to_\varepsilon q_2$)
- ε-closure(q_1) = $\{q_1, q_2\}$
- ε-closure(q_2) = $\{q_2\}$

Step 2: Subset construction (using ε-closures)

Start state of DFA = ε-closure(q_0) = $\{q_0,q_1,q_2\}$ = A. Since it contains q_2, A is also final.

- A = $\{q_0,q_1,q_2\}$, on 0: $\delta(q_0,0) \cup \delta(q_1,0) \cup \delta(q_2,0) = \Phi \cup \{q_1\} \cup \{q_2\} = \{q_1,q_2\}$; ε-closure = $\{q_1,q_2\}$ = B
- A, on 1: $\{q_0\} \cup \Phi \cup \{q_2\} = \{q_0,q_2\}$; ε-closure = $\{q_0,q_1,q_2\}$ = A
- B = $\{q_1,q_2\}$, on 0: $\{q_1\} \cup \{q_2\} = \{q_1,q_2\}$; closure = $\{q_1,q_2\}$ = B
- B, on 1: $\Phi \cup \{q_2\} = \{q_2\}$; closure = $\{q_2\}$ = C
- C = $\{q_2\}$, on 0: $\{q_2\}$ = C; on 1: $\{q_2\}$ = C

Step 3: DFA transition table

DFA state	0	1
→ * A = $\{q_0,q_1,q_2\}$	B	A
* B = $\{q_1,q_2\}$	B	C
* C = $\{q_2\}$	C	C

→ marks the initial state, * marks final states (every DFA state contains q_2, so all are final).

Step 4: Equivalent DFA

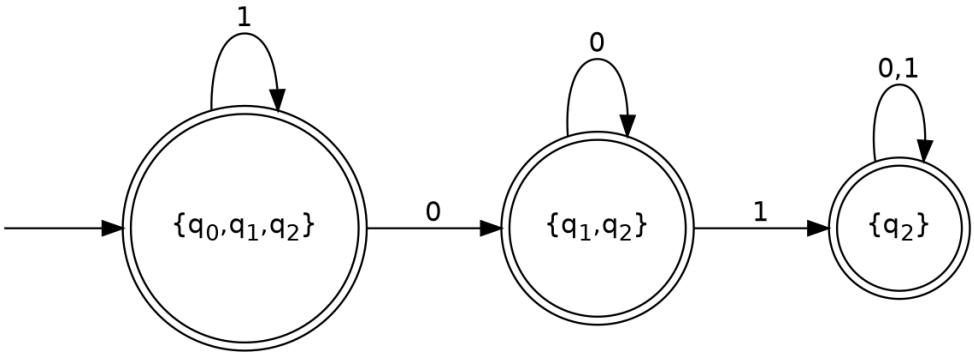


Fig. 5.2 - Equivalent DFA (initial: A, final: A, B, C)

Observation: the language is 1*0*(0+1)* = Σ*, i.e. every binary string is accepted (including the empty string, because the start state is final).